

Raj Aaryan

Software Development Engineer — Backend Systems, Distributed Microservices, Java

Mumbai, India | Open to Relocation

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Work Experience

Jio Platforms Limited

Jul 2022 – Present

Software Development Engineer

- Owned 4 core order-lifecycle microservices (creation, orchestration, fulfillment, reporting) in Java 17 and Spring Boot, processing 20K+ orders/day across 20+ downstream integrations within a 71-engineer team.
- Spearheaded Camunda BPM automation of the order orchestration pipeline, reducing manual intervention by 40% and order processing time from hours to minutes (97% reduction).
- Built a fallout management framework to proactively detect, track, and auto-resolve upstream failures, reducing processing failures by 35% and improving system reliability.
- Integrated Kafka for asynchronous inter-service communication and deployed centralized observability (Elasticsearch, Kibana, Kubernetes) across 10+ microservices, reducing mean time to detection by 30%.
- Built reporting and analytics APIs using MariaDB JSON functions, exposing 50+ configurable fields and reducing manual data extraction effort by 80%.

Samsung Electronics

Aug 2021 – Jun 2022

Software Engineer Intern

- Developed embedded C++ firmware for NAND Flash memory devices, including Bad Block Management (BBM) and encryption/decryption modules.
- Reduced BBM handling latency by 20% and encryption overhead by 15% while maintaining compliance with security requirements.

Projects

E-commerce Order Processing System

Java 17, Spring Boot 3, Kafka, MariaDB, Docker [GitHub]

- Built a production-style order microservice with a 6-state lifecycle, transactional outbox pattern, and idempotent APIs for safe retries under partial failures.
- Implemented inventory reservation with pessimistic row locking and Kafka event publishing with at-least-once delivery guarantees.
- Shipped Docker Compose deployment, Testcontainers integration tests, and a GitHub Actions CI pipeline for automated build and verification.

Distributed In-Memory Cache (Dynamo-Inspired)

Java 17, Distributed Systems, REST API [GitHub]

- Architected a Dynamo-inspired distributed cache with consistent hashing (150 virtual nodes/node) and quorum replication ($N=3$, $R=2$, $W=2$) for fault-tolerant reads and writes.
- Implemented vector clocks, hinted handoff, and read repair to handle node failures and surface concurrent write conflicts.
- Developed pluggable LRU/LFU eviction policies and TTL-based expiration with $O(1)$ cache operations, exposed via a REST API.

Skills

- Languages:** Java, C++, Python, JavaScript
- Backend:** Spring Boot, REST APIs, Kafka, Camunda, Microservices, OpenAPI, Hibernate
- Data & Infra:** MySQL, MariaDB, DynamoDB, Redis, Elasticsearch, Kubernetes, Docker
- Practices:** System Design (LLD/HLD), CI/CD, JUnit, Mockito, Git

Education

BITS Pilani (Pilani Campus)

2017 – 2022

Dual Degree: B.E. in Electronics & Instrumentation, M.Sc. in Mathematics

Achievements

- Received MCN Scholarship for 10 consecutive semesters for academic excellence.
- Received A* rating, Jio Platforms' highest performance rating awarded to the top 1% employees, for 3 consecutive years.
- Mentored an intern on microservices architecture and production onboarding for order-processing services.